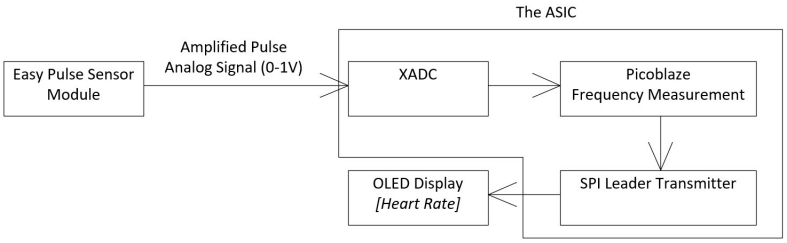
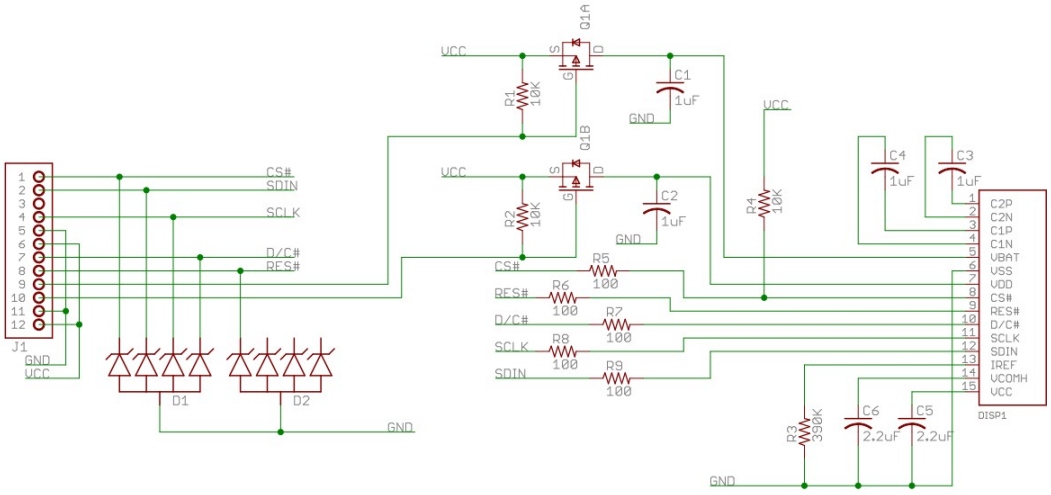
**Spring 2026 CE433 Course Project: An Pulse Sensor ASIC Design**

The goal of this projet is to design a pulse sensor ASIC that includes an off-chip amplifier/filter circuit, an on-chip ADC, Picoblaze CPU, and a SPI modul communicates with an OLED display module (Pmod OLED: 128 x 32 Pixel Monochromatic OLED Display).

The product diagram can be found below.




The Pmod OLED schematic can be found below.




The pinout table can be found below. The CS, MOSI, and SCK pins are standard SPI pins. The fourth pin is D/C which is for Data/Command Control.

Pinout Description Table

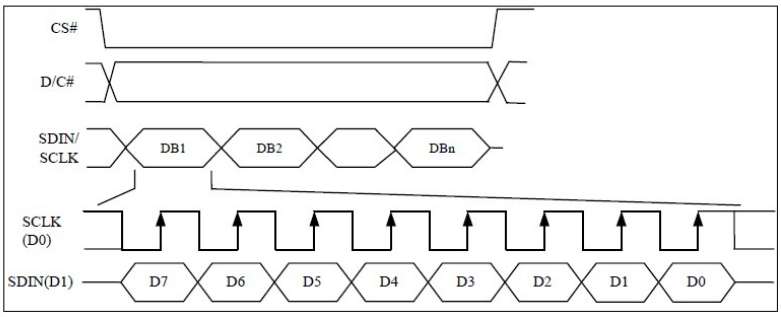
| Header J1 | | | | | |
|---|---|---|---|---|---|
| Pin | Signal | Description | Pin | Signal | Description |
| 1 | CS | Chip Select | 7 | D/C | Data/Command Control |
| 2 | MOSI | Master-Out-Slave-In | 8 | RES | Power Reset |
| 3 | NC | Not Connected | 9 | VBATC | Vbat Battery Voltage Control |
| 4 | SCK | Serial Clock | 10 | VDDC | Vdd Logic Voltage Control |
| 5 | GND | Power Supply Ground | 11 | GND | Power Supply Ground |
| 6 | VCC | Power Supply (3.3V) | 12 | VCC | Power Supply (3.3V) |

The D/C pin determines if the bytes are for the display or for the commands. The SSD1306 datasheet shows the details.
From the datasheet, Section 8.1.3 MCU Serial Interface (4-wire SPI) describes how to use the fourth pin D/C.
SDIN is shifted into an 8-bit shift register on every rising edge of SCLK in the order of D7, D6, ... D0. D/C# is sampled on every eighth clock and the da shift register is written to the Graphic Display Data RAM (GDDRAM) or command register in the same clock.

**Table 8-4 : Control pins of 4-wire Serial interface**

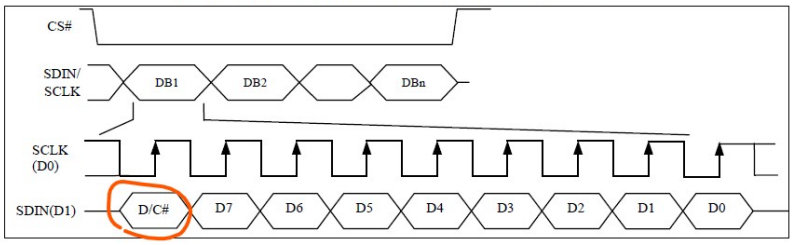
| Function | E(RD#) | R/W#(WR#) | CS# | D/C# | D0 |
|---|---|---|---|---|---|
| Write command | Tie LOW | Tie LOW | L | L | ↑ |
| Write data | Tie LOW | Tie LOW | L | H | ↑ |

**Figure 8-5 : Write procedure in 4-wire Serial interface mode**




However, for the Section 8.1.4 MCU Serial Interface (3-wire SPI), it requires the D/C command to be sent along with data as D8. This requires a differe driver in hardware and software to handle this bit.

**Figure 8-6 : Write procedure in 3-wire Serial interface mode**

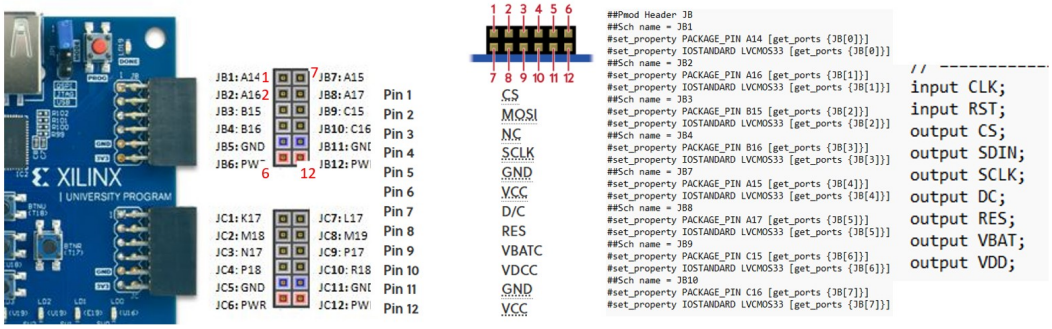



The Pmod OLED example page provided an example package for the Nexys 3 board. We are using the Basys 3 board so some pins, connections, and the co need changes to be compatible with the board.
Open the top file of the design, the port declaration tells that DC is a fourth pin so it must use the 4-wire SPI timing diagram.

```
//////////////////////////////////////////////////////////////////////////////////
module PmodOLEDCtrl(
                CLK,
                RST,
                CS,
                SDIN,
                SCLK,
                DC,
                RES,
                VBAT,
                VDD
    );

        // ===================================================================
        //                                                                Port
Declarations
        // ===================================================================
        input CLK;
        input RST;
        output CS;
        output SDIN;
        output SCLK;
        output DC;
        output RES;
        output VBAT;
        output VDD;
```

Bring the Picoblaze core to this design and write an assembly code to measure the digitized DO from the pulse sensor and count an average heart rate an the OLED display module.

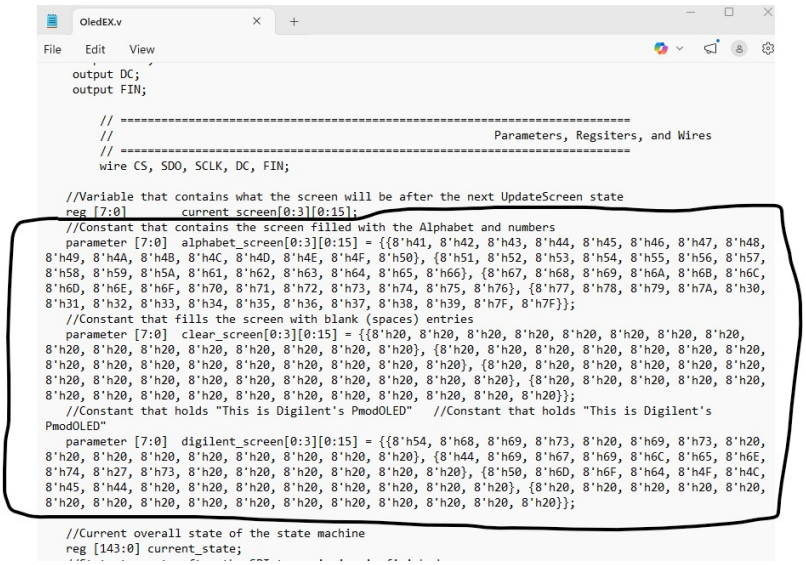
The pins of the PmodOLED is mapped to the JB pins as follows:




```
5
6  # Clock signal
7  set_property PACKAGE_PIN W5 [get_ports CLK]
8  set_property IOSTANDARD LVCMOS33 [get_ports CLK]
9  create_clock -period 10.000 -name sys_clk_pin -waveform {0.000 5.000} -add [get_ports CLK]
10
11 ## Switches
12 set_property PACKAGE_PIN V17 [get_ports {RST}]
13 set_property IOSTANDARD LVCMOS33 [get_ports {RST}]
14 set_property PACKAGE_PIN V16 [get_ports {sw[1]}]
15 set_property IOSTANDARD LVCMOS33 [get_ports {sw[1]}]
16 set_property PACKAGE_PIN W16 [get_ports {sw[2]}]
17 set_property IOSTANDARD LVCMOS33 [get_ports {sw[2]}]
```

```
---
152 #Pmod Header JB
153 #Sch name = JB1
154 set_property PACKAGE_PIN A14 [get_ports {CS}]
155 set_property IOSTANDARD LVCMOS33 [get_ports {CS}]
156 #Sch name = JB2
157 set_property PACKAGE_PIN A16 [get_ports {SDIN}]
158 set_property IOSTANDARD LVCMOS33 [get_ports {SDIN}]
159 #Sch name = JB3
160 set_property PACKAGE_PIN B15 [get_ports {JB[2]}]
161 set_property IOSTANDARD LVCMOS33 [get_ports {JB[2]}]
162 #Sch name = JB4
163 set_property PACKAGE_PIN B16 [get_ports {SCLK}]
164 set_property IOSTANDARD LVCMOS33 [get_ports {SCLK}]
165 #Sch name = JB7
166 set_property PACKAGE_PIN A15 [get_ports {DC}]
167 set_property IOSTANDARD LVCMOS33 [get_ports {DC}]
168 #Sch name = JB8
169 set_property PACKAGE_PIN A17 [get_ports {RES}]
170 set_property IOSTANDARD LVCMOS33 [get_ports {RES}]
171 #Sch name = JB9
172 set_property PACKAGE_PIN C15 [get_ports {VBAT}]
173 set_property IOSTANDARD LVCMOS33 [get_ports {VBAT}]
174 #Sch name = JB10
175 set_property PACKAGE_PIN C16 [get_ports {VDD}]
176 set_property IOSTANDARD LVCMOS33 [get_ports {VDD}]
177
```

The following content in OledEX.v must be replaced by the initial begin/end block listed below since the format is not supported by Vivado.




initial begin

// ================= Alphabet Screen =================

// Row 0

alphabet_screen[0][0]=8'h41; alphabet_screen[0][1]=8'h42;

alphabet_screen[0][2]=8'h43; alphabet_screen[0][3]=8'h44;

alphabet_screen[0][4]=8'h45; alphabet_screen[0][5]=8'h46;

alphabet_screen[0][6]=8'h47; alphabet_screen[0][7]=8'h48;

alphabet_screen[0][8]=8'h49; alphabet_screen[0][9]=8'h4A;

alphabet_screen[0][10]=8'h4B; alphabet_screen[0][11]=8'h4C;

alphabet_screen[0][12]=8'h4D; alphabet_screen[0][13]=8'h4E;

alphabet_screen[0][14]=8'h4F; alphabet_screen[0][15]=8'h50;

// Row 1

alphabet_screen[1][0]=8'h51; alphabet_screen[1][1]=8'h52;

alphabet_screen[1][2]=8'h53; alphabet_screen[1][3]=8'h54;

alphabet_screen[1][4]=8'h55; alphabet_screen[1][5]=8'h56;

alphabet_screen[1][6]=8'h57; alphabet_screen[1][7]=8'h58;

alphabet_screen[1][8]=8'h59; alphabet_screen[1][9]=8'h5A;

alphabet_screen[1][10]=8'h61; alphabet_screen[1][11]=8'h62;

alphabet_screen[1][12]=8'h63; alphabet_screen[1][13]=8'h64;

alphabet_screen[1][14]=8'h65; alphabet_screen[1][15]=8'h66;

// Row 2

alphabet_screen[2][0]=8'h67; alphabet_screen[2][1]=8'h68;

alphabet_screen[2][2]=8'h69; alphabet_screen[2][3]=8'h6A;

alphabet_screen[2][4]=8'h6B; alphabet_screen[2][5]=8'h6C;

alphabet_screen[2][6]=8'h6D; alphabet_screen[2][7]=8'h6E;

alphabet_screen[2][8]=8'h6F; alphabet_screen[2][9]=8'h70;

alphabet_screen[2][10]=8'h71; alphabet_screen[2][11]=8'h72;

alphabet_screen[2][12]=8'h73; alphabet_screen[2][13]=8'h74;

alphabet_screen[2][14]=8'h75; alphabet_screen[2][15]=8'h76;

// Row 3

alphabet_screen[3][0]=8'h77; alphabet_screen[3][1]=8'h78;

alphabet_screen[3][2]=8'h79; alphabet_screen[3][3]=8'h7A;

alphabet_screen[3][4]=8'h30; alphabet_screen[3][5]=8'h31;

alphabet_screen[3][6]=8'h32; alphabet_screen[3][7]=8'h33;

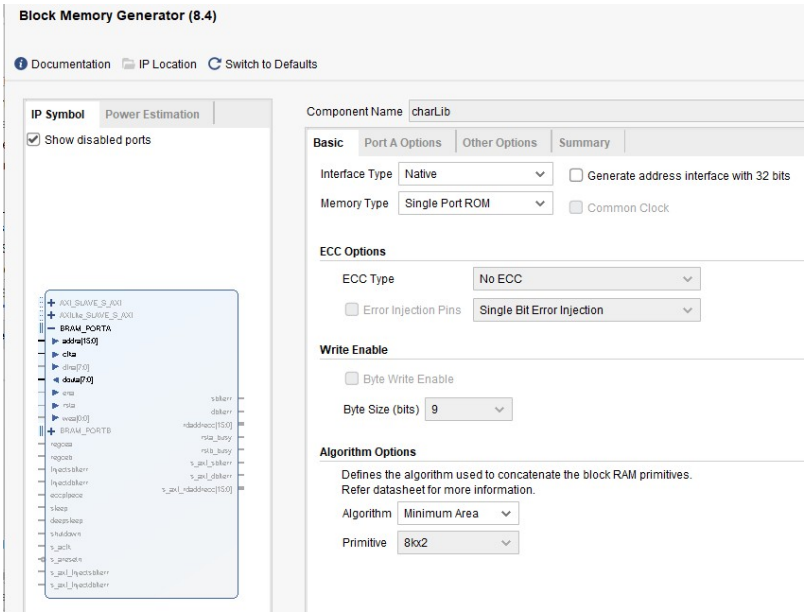
alphabet_screen[3][8]=8'h34; alphabet_screen[3][9]=8'h35;

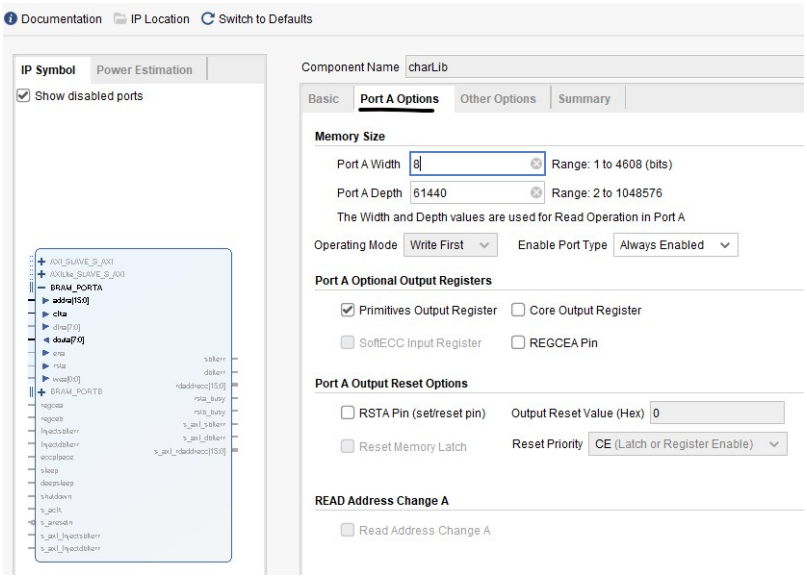
alphabet_screen[3][10]=8'h36; alphabet_screen[3][11]=8'h37;

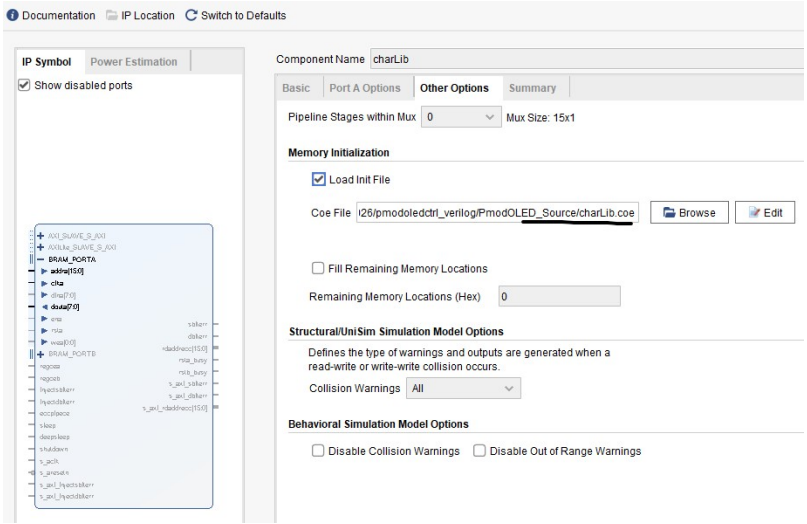
alphabet_screen[3][12]=8'h38; alphabet_screen[3][13]=8'h39;

```
    alphabet_screen[3][14]=8'h7F; alphabet_screen[3][15]=8'h7F;

    // ================= Clear Screen =================
    for (r = 0; r < 4; r = r + 1)
       for (c = 0; c < 16; c = c + 1)
          clear_screen[r][c] = 8'h20;

    // ================= Digilent Screen =================
    // Row 0: "This is"
    digilent_screen[0][0]=8'h54; digilent_screen[0][1]=8'h68;
    digilent_screen[0][2]=8'h69; digilent_screen[0][3]=8'h73;
    digilent_screen[0][4]=8'h20; digilent_screen[0][5]=8'h69;
    digilent_screen[0][6]=8'h73; digilent_screen[0][7]=8'h20;
    digilent_screen[0][8]=8'h20; digilent_screen[0][9]=8'h20;
    digilent_screen[0][10]=8'h20; digilent_screen[0][11]=8'h20;
    digilent_screen[0][12]=8'h20; digilent_screen[0][13]=8'h20;
    digilent_screen[0][14]=8'h20; digilent_screen[0][15]=8'h20;

    // Row 1: "Digilent's"
    digilent_screen[1][0]=8'h44; digilent_screen[1][1]=8'h69;
    digilent_screen[1][2]=8'h67; digilent_screen[1][3]=8'h69;
    digilent_screen[1][4]=8'h6C; digilent_screen[1][5]=8'h65;
    digilent_screen[1][6]=8'h6E; digilent_screen[1][7]=8'h74;
    digilent_screen[1][8]=8'h27; digilent_screen[1][9]=8'h73;
    digilent_screen[1][10]=8'h20; digilent_screen[1][11]=8'h20;
    digilent_screen[1][12]=8'h20; digilent_screen[1][13]=8'h20;
    digilent_screen[1][14]=8'h20; digilent_screen[1][15]=8'h20;

    // Row 2: "PmodOLED"
    digilent_screen[2][0]=8'h50; digilent_screen[2][1]=8'h6D;
    digilent_screen[2][2]=8'h6F; digilent_screen[2][3]=8'h64;
    digilent_screen[2][4]=8'h4F; digilent_screen[2][5]=8'h4C;
    digilent_screen[2][6]=8'h45; digilent_screen[2][7]=8'h44;
    digilent_screen[2][8]=8'h20; digilent_screen[2][9]=8'h20;
    digilent_screen[2][10]=8'h20; digilent_screen[2][11]=8'h20;
    digilent_screen[2][12]=8'h20; digilent_screen[2][13]=8'h20;
    digilent_screen[2][14]=8'h20; digilent_screen[2][15]=8'h20;

    // Row 3: blank
    for (c = 0; c < 16; c = c + 1)
       digilent_screen[3][c] = 8'h20;
end
```

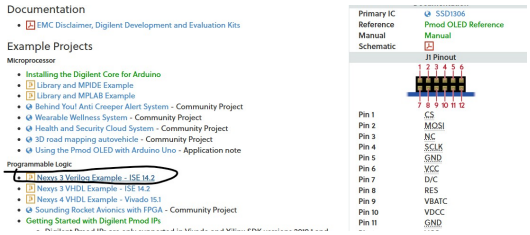
In addition to the modifications above, you also need to create a memory 'charLib' to store the charLib.coe data. You can tell that charLib is instantiatec
module

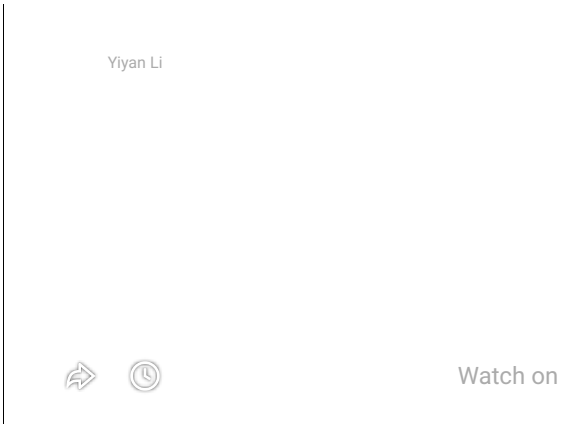

Block Memory Generator (8.4)
Documentation
IP Location
Switch to Defaults
IP Symbol
Power Estimation
Show disabled ports
Component Name
charLib
Basic
Port A Options
Other Options
Summary
Interface Type
Native
Generate address interface with 32 bits
Memory Type
Single Port ROM
Common Clock
ECC Options
ECC Type
No ECC
Error Injection Pins
Single Bit Error Injection
Write Enable
Byte Write Enable
Byte Size (bits)
9
Algorithm Options
Defines the algorithm used to concatenate the block RAM primitives.
Refer datasheet for more information.
Algorithm
Minimum Area
Primitive
8kx2


Documentation
IP Location
Switch to Defaults
IP Symbol
Power Estimation
Show disabled ports
Component Name
charLib
Basic
Port A Options
Other Options
Summary
Memory Size
Port A Width
8
Range: 1 to 4608 (bits)
Port A Depth
61440
Range: 2 to 1048576
The Width and Depth values are used for Read Operation in Port A
Operating Mode
Write First
Enable Port Type
Always Enabled
Port A Optional Output Registers
Primitives Output Register
Core Output Register
SoftECC Input Register
REGCEA Pin
Port A Output Reset Options
RSTA Pin (set/reset pin)
Output Reset Value (Hex)
0
Reset Memory Latch
Reset Priority
CE (Latch or Register Enable)
READ Address Change A
Read Address Change A


Documentation
IP Location
Switch to Defaults
IP Symbol
Power Estimation
Show disabled ports
Component Name
charLib
Basic
Port A Options
Other Options
Summary
Pipeline Stages within Mux
0
Mux Size: 15x1
Memory Initialization
Load Init File
Coe File
l26/pmodoledctrl_verilog/PmodOLED_Source/charLib.coe
Browse
Edit
Fill Remaining Memory Locations
Remaining Memory Locations (Hex)
0
Structural/UniSim Simulation Model Options
Defines the type of warnings and outputs are generated when a
read-write or write-write collision occurs.
Collision Warnings
All
Behavioral Simulation Model Options
Disable Collision Warnings
Disable Out of Range Warnings

Here is the result of the example from Digilent.




**Tasks:**
Task 1: Turn the PmodOLEDCtrl.v and the submodules into a Basys3 compatible design and display any characters on the OLED to prove its functionality. Picoblaze involved, only use the code from the example and make it Basys3 compatible. (50 points)
Task 2: Add Picoblaze to the design, probe A0, read the XADC's output and display the binary form on the on-board LEDs. (50 points)
Task 3: Use assembly and Picoblaze for frequency measurement and display heart rate on the OLED and at the same time, display realtime waveforms on plot. (50 points)

References:
1. Pmod OLED product page
2. Pmod OLED support page
3. Pmod OLED examples page
3. SSD1306 datasheet